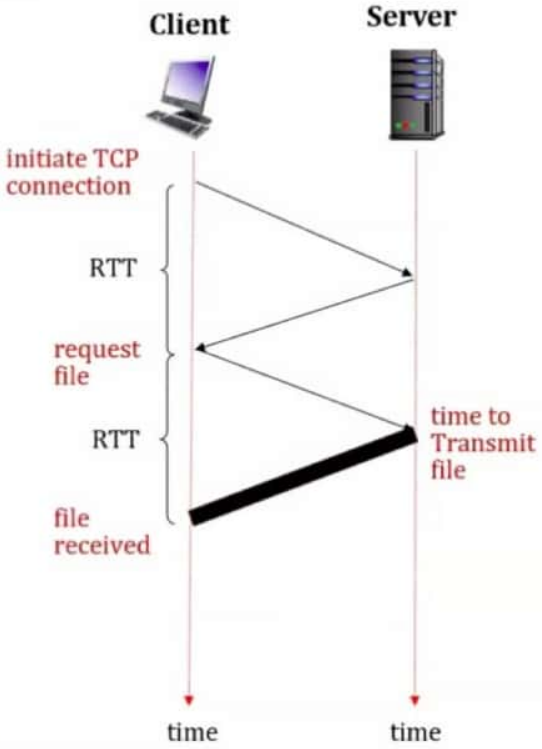
HTTP Connection

- There are two types of HTTP Connection:
  1. Non-Persistent HTTP
  2. Persistent HTTP

- RTT(round-trip time): A time for a small packet to travel from client to server and server to client.

- HTTP response time:
  - 1-RTT to initiate TCP connection.
  - 1-RTT for HTTP request & response.
  - File transmission time



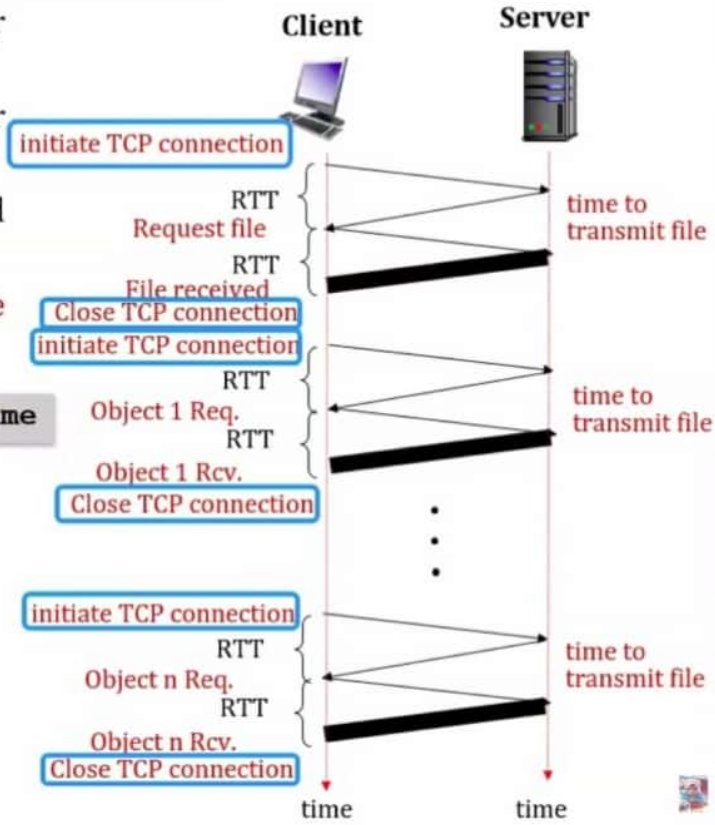

```
HTTP Response Time = No. of RTT + file transmit time
```

Scanned with CamScanner

Non Persistent HTTP

- Non-persistent connections are the default mode for HTTP/1.0.
- A non-persistent connection is closed after the server sends the requested object to the client.
- The connection is used exactly for one request and one response.
- For downloading multiple objects it required multiple connections.

```
Non-Persistent HTTP = 2 RTT + 1 file transmit time
```



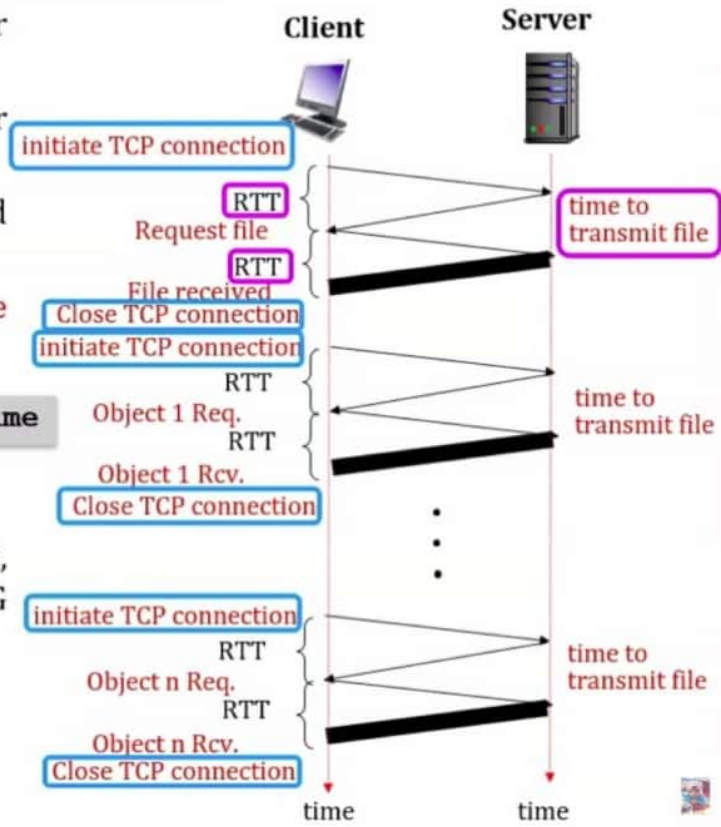
Scanned with CamScanner

Non Persistent HTTP

- Non-persistent connections are the default mode for HTTP/1.0.
- A non-persistent connection is closed after the server sends the requested object to the client.
- The connection is used exactly for one request and one response.
- For downloading multiple objects it required multiple connections.

```
Non-Persistent HTTP = 2 RTT + 1 file transmit time
```

- Example:
- Transferring a webpage from server to client, webpage consists of a base HTML file and 10 JPEG images.



Scanned with CamScanner

Sample problem

Suppose 10 images need to be downloaded from the HTTP server. The total time taken to request and download 10 images in a non-persistent and persistent connection is:

Non-persistent

2 RTT (Connection time) + 2 * 10 RTT= $22RTT$

Persistent

2 RTT (Connection time) + 10 RTT= $12RTT$

Scanned with CamScanner